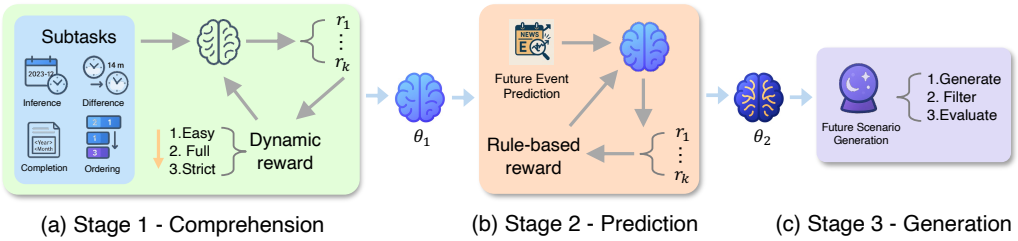

(a) Stage 1 - Comprehension (b) Stage 2 - Prediction (c) Stage 3 - Generation

Figure 2: **Overview of the Time-R1 framework.** The process consists of three stages: (a) Stage 1 establishes foundational understanding by fine-tuning a base LLM on historical data across four temporal subtasks, driven by reinforcement learning (GRPO) and a dynamic reward system, resulting in model θ_1. (b) Stage 2 trains θ_1 for future event time prediction using post-cutoff data and a rule-based reward, producing θ_2. (c) Stage 3 leverages θ_2 for inference-based creative future scenario generation, followed by evaluation, without further RL.

Structured Generation Process. To facilitate complex reasoning, interpretability and structured output, we guide the model generation process. For all tasks, the LLM is prompted using specific templates incorporating system instructions (*i.e.*, instructing the model to reason first: "You are a helpful assistant. You first think about the reasoning process in your mind and then provide the user with the answer.") to generate its reasoning within "<think>...</think>" tags, followed by the final answer within "<answer>...</answer>" tags. The entire generated sequence y, encompassing both thought and answer components, constitutes the output evaluated by the environment.

Policy Optimization using GRPO. The environment evaluates the output y using a task-specific dynamic reward function $R(x, y)$ (detailed in Section 3.3). To optimize the policy parameters θ, we utilize Group Relative Policy Optimization (GRPO) [19]. A key challenge in RL fine-tuning of LLMs is the high variance often associated with policy gradient estimates[35]. GRPO addresses this by calculating the advantage of a generated response relative to other responses sampled for the same input prompt, thereby providing a more stable learning signal without requiring an auxiliary value function.

Specifically, for a given prompt x, we first sample a batch of K responses $\{y_k\}_{k=1}^{K}$ using a reference policy π_{ref} (typically the policy before the update step). After computing the reward $R(x, y_k)$ for each response, the group-normalized advantage $\hat{A}(x, y_k)$ for response y_k is calculated as:

$$\hat{A}(x, y_k) = R(x, y_k) - b(x), \quad \text{where } b(x) = \frac{1}{K}\sum_{j=1}^{K} R(x, y_j). \tag{1}$$

This advantage estimate $\hat{A}(x, y_k)$ reflects the relative quality of response y_k compared to the average performance within its group.

To update the policy π_θ stably using this advantage, we employ a clipped surrogate objective function, similar in structure to that used in PPO [18], which helps prevent large, detrimental policy updates. Let the probability ratio be $r_k(\theta) = \frac{\pi_\theta(y_k|x)}{\pi_{\text{ref}}(y_k|x)}$. The per-sample clipped objective term is:

$$L_k^{\text{CLIP}}(\theta) = \min\Big(r_k(\theta)\hat{A}(x, y_k),\ \text{clip}\left(r_k(\theta), 1-\epsilon, 1+\epsilon\right)\hat{A}(x, y_k)\Big) \tag{2}$$

where ϵ is the clipping hyperparameter. The overall objective function $J_{\text{GRPO}}(\theta)$ maximized during training balances the expected clipped advantage with a KL-divergence penalty against the reference policy π_{ref}:

$$\max_\theta J_{\text{GRPO}}(\theta) = \mathbb{E}_{x\sim\mathcal{D},\{y_k\}\sim\pi_{\text{ref}}}[\frac{1}{K}\sum_{k=1}^{K} L_k^{\text{CLIP}}(\theta)] \ - \ \beta\,\mathbb{E}_{x\sim\mathcal{D}}\mathbb{D}_{\text{KL}}[\pi_\theta(\cdot \mid x) \parallel \pi_{\text{ref}}(\cdot \mid x)], \tag{3}$$

where $\mathcal{D}$ is the training dataset union, β controls the KL penalty strength, $\mathbb{D}_{\text{KL}}$ is the Kullback–Leibler divergence, and π_{ref} is the stage-specific frozen reference policy (initialized from Qwen2.5-3B-Instruct for Stage 1) used for both advantage calculation reference and KL regularization. This objective guides the policy towards higher rewards, leveraging the stable GRPO advantage estimates within a constrained optimization framework.

3.2 Time-R1: A Three-Stage Temporal Learning Framework

To empirically evaluate the effectiveness of our proposed methodology (outlined in Section 3.1), we designed a comprehensive three-stage experimental procedure to train **Time-R1**, as shown in Figure 2. This staged approach aims to progressively cultivate sophisticated temporal logic, prediction, and generation capabilities within the Large Language Model (LLM). We detail each stage below.

3.2.1 Stage 1 - Comprehension: Foundational Temporal Understanding via RL Fine-tuning

Objective. The primary goal of this initial stage is to establish a robust foundation for temporal comprehension within the LLM. We aim to instill the ability to interpret fundamental temporal relationships between events and their corresponding times by fine-tuning the model using historical news data from *before* its knowledge cutoff date.

Dataset. We construct a specialized dataset derived from a large corpus of New York Times (NYT) news articles [36] (over 200,000) spanning eight years, from January 2016 to December 2023. We extract the headline h and abstract a of the news article to represent each event E, *i.e.*, $E = (h, a)$. Details can be found in Appendix B.1.

Subtasks. From this corpus, we curate data instances tailored to four specific and fundamental temporally-focused and logic-based subtasks [37, 38]: **(1) Timestamp Inference:** Infer the specific date t (*e.g.*, 2023-12) associated with a described event E. **(2) Time-Difference Estimation:** Estimate the temporal gap Δt (*e.g.*, 14 months) between two described events, E_1 and E_2. **(3) Event Ordering:** Determine the correct chronological sequence C (*e.g.*, Event order: 2-1-3) of three events E_1, E_2 and E_3 presented out of order. **(4) Masked Time Entity Completion:** Fill in a masked temporal expression M_e (*i.e.*, <Year> and <Month>) within a given event description E'.

In order to help the model develop general logic and indeed acquire the skill to accurately map events to their respective times from textual clues, we force the model to infer each event's date first and then give a task-specific answer for every subtask except the first. Both would be judged a score that would then serve as a part of the reward (see Section 3.3). Consequently, this prevents the model from merely guessing the final answer implicitly. For instance, for the Masked Time Entity Completion task, success hinges on the model's ability to discern detailed semantics from the surrounding text. This is crucial because the specific temporal entity to be completed often refers to a time distinct from the primary date of the event itself, thus pushing the model beyond simple date extraction towards a deeper contextual understanding to answer both correctly. By mastering these diverse subtasks, the LLM (*i.e.*, a model checkpoint, denoted θ_1) builds a robust foundational temporal understanding.

3.2.2 Stage 2 - Prediction: Future Event Time Prediction via RL Fine-tuning

Objective. After obtaining the foundational capabilities developed in Stage 1, the objective of Stage 2 is to further train the model to predict the timing of future events occurring *after* its initial knowledge cutoff (2023). This involves teaching the model to recall relevant and similar events in the past and their occurrence dates, extrapolate learned temporal development patterns and anticipate future event occurrences based on emerging, post-cutoff information.

Dataset. For Stage 2, the training dataset, denoted $\mathcal{D}^{(2)}_{\text{train}}$, is meticulously constructed to facilitate fair evaluation and strictly prevent data leakage from the test period. To ensure a level playing field and align with the knowledge cutoff of the latest baseline models (e.g., DeepSeek-V3-0324-671B with a knowledge cutoff in July 2024), we first incorporate real news data. Specifically, we include a corpus of 7,000 real news articles from January 2024 to July 2024. To train for predicting events beyond this cutoff (August 2024 - February 2025) without using real data from this period, we employ a data synthesis strategy. The synthetic dataset, created using the DeepSeek-V3 model informed by news from May to July 2024, constitutes approximately only half the volume of the real news data used for the earlier months. This approach of using exclusively synthetic data for the future period is a deliberate measure to strictly avoid any potential data leakage, as the test dataset $\mathcal{D}^{(2)}_{\text{test}}$ is real news events from this period (August 2024 - February 2025). Further details about the datasets can be found in Appendix B.2.

Task. In this stage, the model predicts the specific date t for a news event E based on its extracted headline h and abstract a.

Initializing the model with the checkpoint θ_1 obtained from Stage 1, we continue the fine-tuning process using GRPO on post-cutoff news while carefully controlling the information availability to simulate a true "future prediction" scenario. After training, this stage addresses the challenge that LLMs normally cannot generalize to events post-training [39] and results in another model checkpoint, θ_2, specialized in future event time prediction.

3.2.3 Stage 3 - Generation: Creative Future Scenario Generation and Evaluation

Objective. In the final stage, we pivot from training to application – aiming to leverage the logical and predictive capabilities instilled in Stages 1 and 2 to enable the fine-tuned model to directly generate plausible, diverse, and temporally coherent future scenarios. This moves beyond predicting specific event times to creatively generating descriptions of hypothetical events given a specific future date.

Methodology. This stage utilizes the model checkpoint θ_2, obtained from Stage 2, exclusively for inference without any further RL fine-tuning. The process involves three sequential steps: future news generation, diversity-based filtering, and plausibility evaluation against real news.

First, the model generates hypothesized news events for specified future months M (*i.e.*, July 2024 onwards). To ensure comprehensive topical coverage, generation is conditioned on $T = 8$ common and distinct themes τ (*e.g.*, Foreign Affairs, Business, Technology, Politics). To enhance the richness of the output pool, each prompt asks the model to create multiple unique news (*i.e.*, 3). This process results in a raw set of generated news items $\mathcal{G}_{\text{raw}}$ including each month m and theme τ.

Second, to curate a varied and non-redundant set of scenarios for evaluation, a diversity filtering process is applied to the raw generated articles $\mathcal{G}_{\text{raw}}$. We compute semantic embeddings $\mathbf{g} \in \mathbb{R}^{384}$ for each generated item g using all-MiniLM-L6-v2 encoder [40], which retains excellent semantic capture capabilities through knowledge distillation from larger models [41]. Within each theme τ and month m, a greedy selection algorithm iteratively constructs a diverse subset. This filtering yields a curated set $\mathcal{G}_{\text{filt},m}$ containing $N_{\text{div}} = 5$ high-diversity news items per theme per month, totaling $N_g = T \times N_{\text{div}} = 40$ representative generated scenarios for each month m.

Finally, the realism and plausibility of the generated future scenarios are quantified through comparison with actual news events from the corresponding future months. The ground truth consists of real news events r from the held-out test dataset $\mathcal{D}^{(2)}_{\text{test}}$, partitioned by month m into sets $\mathcal{D}_{\text{real},m}$. We compute semantic embeddings $\mathbf{A}_g$ for the filtered generated news items $g \in \mathcal{G}_{\text{filt},m}$ and $\mathbf{B}_r$ for the real news items $r \in \mathcal{D}_{\text{real},m}$, using the same "all-MiniLM-L6-v2" model. The semantic relatedness between a generated item $\mathbf{A}_g$ and a real item $\mathbf{B}_r$ is measured using cosine similarity: $\text{sim}(\mathbf{A}_g, \mathbf{B}_r) = \cos(\phi) = \frac{\mathbf{A}_g \cdot \mathbf{B}_r}{\|\mathbf{A}_g\| \|\mathbf{B}_r\|}$, where ϕ represents the angle between the 384-dimensional embedding vectors. To assess overall plausibility for a given month m, we calculate the Average Maximum Similarity (AvgMaxSim) score. For each generated news item $\mathbf{A}_{g,i}$ ($i = 1, \ldots, N_g$), we find its maximum similarity to any real news item in that month, $\max_{\mathbf{B}_r \in \mathcal{D}_{\text{real},m}} \text{sim}(\mathbf{A}_{g,i}, \mathbf{B}_r)$. The AvgMaxSim score is the average of these maximum similarity values across all N_g generated items:

$$\text{AvgMaxSim}_m = \frac{1}{N_g} \sum_{i=1}^{N_g} \left(\max_{\mathbf{B}_r \in \mathcal{D}_{\text{real},m}} \text{sim}(\mathbf{A}_{g,i}, \mathbf{B}_r) \right) \tag{4}$$

This metric quantifies, on average, how closely the generated plausible future events align semantically with events that actually transpired during that month. The process culminates in generating monthly AvgMaxSim reports and visualizations, facilitating quantitative comparisons against baseline generative models or ablations of our framework.

In summary, Stage 3 serves as evidence for the generalization fostered by our first two stages RL framework. It reveals that the strong temporal grounding comprehension and predictive skills learned previously, combined with the LLM's innate linguistic abilities, readily and effectively generalize, allowing the model to anticipate future event dynamics and generate plausible, creative scenarios accordingly, without task-specific fine-tuning for this generative capability.

3.3 Reward Design

A meticulously engineered reward function, $R(x, y)$, underpins the success of our Time-R1 framework. Its comprehensive and rigorous design, refined through iterative experimentation, has proven